

Electric & Hybrid-Electric Aviation Engineering

Subdomain Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

What This Document Covers: Electric and hybrid-electric aviation is the fastest-restructuring subdomain in aerospace engineering. What was, in 2015, a collection of academic research programmes and ambitious startups has become, in 2025, a sector with dozens of certificated or near-certification aircraft types, billions of dollars of investment, a published EASA Special Condition for VTOL aircraft (SC-VTOL-01), an FAA powered-lift certification pathway, and the first commercial eVTOL operations beginning in select markets. The engineering challenges that define this sector — packing energy into a structure that must fly, converting it to shaft power through electric motors and inverters with minimum loss, managing the thermal consequences, certifying a novel power architecture under regulations designed for gas turbines, and doing all of this at a weight and cost that produces a viable transport product — are among the most technically demanding in aerospace, and they are largely unsolved at the commercial scale the market requires.

The field sits at the intersection of electrochemistry, power electronics, electric machine design, thermal management, aircraft systems integration, and airworthiness certification. Engineers who are strong in only one of these dimensions are at a disadvantage relative to those who can move fluently across the interfaces between them. The battery pack designer who does not understand the power electronics architecture into which the pack is integrated, the motor designer who does not understand the thermal management system that dissipates the losses from their machine, and the systems engineer who does not understand the electrochemical ageing mechanisms that limit the battery's usable life across a fleet's operational lifetime — all of these are producing incomplete engineering, and in a sector where weight margins are razor-thin and certification evidence is being developed from scratch, incomplete engineering is visible quickly.

The employer landscape spans pure eVTOL developers (Joby Aviation, Lilium — in insolvency as of late 2024 but its technology assets continuing under new ownership, Archer Aviation, Wisk Aero, Overair's technology lineage, Volocopter, EHang, AutoFlight, Vertical Aerospace, Beta Technologies, Supernal, Skyryse), hybrid-electric regional aircraft developers (Heart Aerospace, Harbour Air conversion, Universal Hydrogen, ZeroAvia for hydrogen fuel cell), established OEMs entering the sector (Airbus CityAirbus NextGen, Boeing subsidiary Wisk, Embraer Eve Air Mobility, Rolls-Royce Spirit of Innovation derivative programmes), propulsion system suppliers (Rolls-Royce electrical, GE Aviation electrical systems, Safran Electrical & Power, Collins Aerospace electrical systems, Honeywell electric propulsion), battery suppliers and integrators (Sion Power, CATL, Samsung SDI in aviation-adapted formats, PolyJoule), and research institutions (NASA Electrified Aircraft Propulsion programme, DLR Electric Flight, ONERA MERMOZ, Cranfield University, Delft University of Technology).

PART I: WHAT ELECTRIC & HYBRID-ELECTRIC AVIATION ACTUALLY LOOKS FOR

By Level

Intern / Graduate

At entry level, eVTOL and electric propulsion employers are screening for three things simultaneously: the physical understanding of electrical energy storage and conversion that allows a candidate to reason

quantitatively about what an electric propulsion system can and cannot do; the systems-level thinking that connects the battery, the inverter, the motor, and the airframe into a coherent weight-and-power budget; and the ability to engage with novel technical problems without the safety net of established handbook methods. This last quality is specifically important in electric aviation — where there are no handbook methods for most problems — and it is what most distinguishes this sector from conventional aerospace engineering in its interview culture.

The question that most reliably distinguishes strong from weak graduates is not “what is specific energy?” but “a 4-passenger eVTOL needs to fly a 15-minute mission at 120 kW average power — estimate the battery mass required at three different specific energy levels (150, 200, 250 Wh/kg) and explain what this tells you about the viability of the mission at each level and what proportion of the aircraft’s maximum takeoff weight the battery represents.” The candidate who can work through this — in SI units, with explicit assumptions about depth of discharge, pack-level versus cell-level energy density, and reserve margins — and then reason about what the resulting battery fraction implies for payload and structural weight, is demonstrating the systems-level energy reasoning that is the entry-level competency of electric aviation engineering.

Graduate interviews at eVTOL developers typically move quickly from a take-home or live technical problem to a discussion of the candidate’s experience with real hardware — not necessarily electric aviation hardware, but any system involving batteries, motors, power electronics, or embedded control. The gap between candidates who have built something electrical and those who have only studied it is immediately apparent. Building a battery management system for a student rocketry project, designing a motor controller for a FSAE car, or characterising a lithium-ion cell through a charge-discharge cycle on a cycler — all of these provide the practical electrical engineering intuition that textbook study cannot substitute.

Mid-Level (Engineer, Senior Engineer, ~3–8 years)

At mid-level, electric aviation interviews become specific to the engineering problem the employer is trying to solve. The interviewer is assessing whether you can size a motor for a specified torque and speed requirement while meeting a thermal constraint, design a battery pack thermal management system that keeps cells within temperature limits during a fast-discharge mission, assess whether a proposed inverter topology is appropriate for the motor type and switching frequency requirements, or evaluate whether a proposed battery state-of-health monitoring algorithm is conservative enough to meet the SC-VTOL-01 safety requirements for the loss of propulsion failure condition.

Technical interviews at eVTOL developers are often unusually direct and fast-moving relative to legacy aerospace employers — reflecting the startup culture of most of the sector. The expectation is that the candidate can engage with a real design problem at the first technical meeting, rather than spending weeks in orientation before making any contribution. The mid-level engineer who arrives at a Joby or Archer interview having studied the publicly available documents from those companies’ FAA certification engagement — the G-1 issue papers, the ELOS findings, the means of compliance submissions — is demonstrating the employment-specific engagement that these companies value and that most candidates do not provide.

Senior (Principal Engineer, Technical Lead, Chief Engineer)

At senior level, electric aviation interviews assess the ability to make system-level architectural decisions under conditions of deep technical uncertainty — where the certification basis is still being negotiated with the authority, where the battery technology is still evolving faster than any product development cycle, and where the failure modes of the integrated electrical power system are not fully characterised by any existing test database. The senior electric aviation engineer must be simultaneously technically rigorous (the physics of electrochemical degradation and power electronics switching losses do not have approximate solutions) and comfortable with uncertainty in a way that traditional aerospace engineering rarely requires.

The distinguishing credential at senior level is having taken a complete electrical propulsion system or power architecture — from requirement through design through hardware test through airworthiness demonstration — on a certificated or near-certification aircraft, and having worked through the specific moments where the regulatory methodology had to be developed rather than applied. These moments are specific to this sector and cannot be replicated in any conventional aerospace programme.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

- **Lithium-ion cell electrochemistry** — the mechanisms of energy storage, the C-rate relationship, ageing, and thermal runaway. The lithium-ion cell is the foundational energy storage technology of the electric aviation sector, and understanding it at the level of its electrochemical mechanisms — the intercalation

and de-intercalation of lithium ions between the anode and cathode during charge and discharge, the role of the solid-electrolyte interphase (SEI) in cell ageing, the thermal runaway cascade that begins with lithium plating or separator penetration and progresses through exothermic decomposition of electrolyte and cathode materials — is the electrochemical literacy that separates a battery engineer from an engineer who has used batteries. The C-rate (the ratio of discharge current to the cell's nominal 1-hour capacity) determines both the power delivery capability and the rate of cell heating; understanding why a cell at 4C discharge produces substantially more heat than the same cell at 1C, and what the thermal consequences are for cell temperature uniformity in a large pack, is the practical electrochemistry that battery system designers work with daily.

- **Battery Management System design** — state estimation, cell balancing, thermal management interface, and the safety architecture. The BMS is the electronic system that monitors and controls the battery pack — measuring cell voltages, currents, and temperatures, estimating state of charge and state of health, implementing cell balancing to prevent charge imbalance from limiting pack capacity, and enforcing the protection functions (overvoltage, undervoltage, overcurrent, overtemperature, undertemperature) that prevent cell damage or thermal runaway. State-of-charge estimation — using Coulomb counting, open-circuit voltage correlation, or Kalman filter-based methods that combine both — is a specific algorithmic challenge that is well-studied and consistently tested. The safety architecture of the BMS for an airborne application — specifically the failure mode and effects analysis that demonstrates that no single BMS failure can result in loss of propulsion above the probability allowed for the worst-case failure condition under SC-VTOL-01 — is the certification-relevant BMS engineering that eVTOL developers require at mid-level.
- **Electric motor design and selection** — Permanent Magnet Synchronous Motor (PMSM), Brushless DC (BLDC), induction motor architectures, and the torque-speed-efficiency map. The primary motor type in electric aviation is the PMSM — it offers high power density, high efficiency across a wide speed range, and well-understood control properties. Understanding the fundamental torque production mechanism (Lorentz force on a current-carrying conductor in a magnetic field, implemented through the interaction of the rotating magnetic field of the stator with the permanent magnets of the rotor), why the motor's efficiency peaks at a specific torque and speed combination, and what the thermal limits are — the winding insulation temperature limit that governs continuous operation and the peak temperature that limits transient peak power — is the electric machine fundamentals that motor selection and integration engineers require. The distinction between a PMSM and a switched reluctance motor (SRM) — which uses no permanent magnets, is more fault-tolerant but has higher torque ripple, and requires more complex control — is specifically relevant to safety-critical aviation applications where the rare-earth magnet supply chain risk of PMSMs is a concern.
- **Power electronics** — inverter topology, switching losses, thermal management, and EMI. The inverter converts DC power from the battery to the variable-frequency, variable-voltage AC power required by the electric motor. The standard topology for aviation power electronics is the three-phase two-level voltage source inverter using Silicon Carbide (SiC) MOSFETs — SiC offers substantially lower switching losses at higher switching frequencies than conventional silicon IGBTs, enabling smaller filter inductors and lower EMI at equivalent power levels. Understanding the switching loss mechanism (the product of the switching transient voltage and current during turn-on and turn-off, integrated over the switching event), why switching frequency is a trade between filter size (lower frequency requires larger inductors) and switching losses (higher frequency increases losses), and what the thermal consequences of these losses are for the heat sink and cooling system design are the power electronics fundamentals that every eVTOL electrical system engineer requires. The EMI challenge — the conducted and radiated electromagnetic interference generated by the fast-switching SiC inverter — is a specific and often underestimated certification challenge, because aircraft avionics must continue to function correctly in the electromagnetic environment created by a kilowatt-scale inverter switching at tens of kilohertz.
- **Thermal management** — heat rejection from battery, motor, and inverter during worst-case mission profiles. Electric propulsion systems generate heat at every energy conversion step — from chemical to electrical in the battery (internal resistance losses), from electrical to mechanical in the motor (copper and iron losses), and from electrical to electrical in the inverter (switching and conduction losses). In a conventional aircraft, heat rejection is achieved through the jet exhaust and the aircraft skin; in an

electric aircraft, the battery, motor, and inverter must each have dedicated thermal management systems — typically liquid cooling circuits for the motor and inverter, and either liquid cooling or phase change material for the battery. Understanding how the thermal management system is sized — from the worst-case heat generation at peak power during takeoff, through the transient thermal response of the system, to the coolant temperature rise across the load — and what the failure consequence is if any part of the thermal management system is degraded during flight are the thermal management engineering competencies that every electric propulsion system integrator requires.

- **SC-VTOL-01 certification framework** — failure condition classification, propulsion system integrity requirements, and the means of compliance challenges. EASA's Special Condition for VTOL aircraft (SC-VTOL-01, published 2019 and under continuing development) is the primary certification framework for eVTOL aircraft in Europe. It introduces the concept of the Development Assurance Level (DAL) for non-safety-critical architectures that have inherent redundancy, a failure condition classification that differs from CS-25 in addressing the graceful degradation of multi-motor configurations, and specific requirements for the propulsion system integrity demonstration under CS-VTOL-SC.2305 and CS-VTOL-SC.2500. Understanding what SC-VTOL-01 requires for the loss of a single motor (the aircraft must remain controllable with reduced performance margin), what the qualification approach for a novel battery cell chemistry without decades of in-service failure data looks like, and how the propulsion system FMEA for a distributed electric propulsion configuration differs from that of a conventional twin-engine aircraft are the certification framework competencies that are specifically probed at mid-level and above.
- **Distributed electric propulsion** — the aerodynamic and propulsion integration benefits and the control challenges. Distributed electric propulsion (DEP) — the use of multiple smaller electric motors and propulsors distributed across the airframe rather than one or two large engines — enables aerodynamic-propulsion interactions that are not achievable with conventional propulsion. The propulsive efficiency benefit of very high bypass ratio (from many small, slow-turning propulsors) and the boundary layer ingestion benefit (from propulsors positioned to ingest the wing boundary layer and re-energise the wake) are the aerodynamic efficiency arguments for DEP. The failure consequence benefit — that the loss of one motor in a distributed configuration of eight or more motors produces a small fraction of the total thrust loss that engine failure produces in a conventional twin — is the safety architecture argument. The control challenge — that a distributed motor array must be managed as a combined propulsion and flight control effector, with motor speed differentials providing roll, pitch, and yaw control in the hover regime — is the flight control complexity argument that makes DEP vehicles inherently more complex to certify than conventional aircraft.
- **Energy density versus power density** — the trade-off in battery selection and its mission implications. Not all energy storage requirements in electric aviation are the same: a long-range commuter aircraft needs maximum energy per kilogram (high specific energy), while a rotorcraft in hover needs maximum power per kilogram (high specific power). These requirements are in tension in lithium-ion chemistry: cells optimised for high specific energy (thin electrodes, high energy density active materials) typically have lower specific power than cells optimised for high specific power (thick electrodes, high surface area). Understanding the Ragone plot — which maps specific energy against specific power for different electrochemical storage technologies — and what it implies for the selection of cell chemistry (NMC, LFP, NCA, solid-state) for a specific mission profile is the battery system engineering literacy that distinguishes candidates who understand the physics of energy storage from those who know only the headline Wh/kg figure.
- **Hybrid-electric architectures** — series, parallel, turbo-electric, and the optimal hybrid ratio for a given mission. Pure electric propulsion is currently limited by battery specific energy to short-range missions (approximately 100–200 km depending on payload and performance requirements). Hybrid-electric architectures extend range by combining a gas turbine or fuel cell as the primary energy source with a battery that provides peak power during takeoff and enables energy recovery during descent. The series hybrid (gas turbine drives a generator, which charges the battery, which drives electric motors) decouples the turbine from the propulsors and enables turbine operation at constant optimal efficiency but adds the weight of the generator and power electronics. The parallel hybrid (both the gas turbine and the electric motor can drive the propulsor directly) avoids the series conversion losses but requires mechanical coupling complexity. The optimal hybrid ratio — the fraction of takeoff power provided by

the battery — depends on the mission profile and is specifically an energy-to-power ratio optimisation problem.

- **Hydrogen fuel cell propulsion** — the Proton Exchange Membrane Fuel Cell (PEMFC) operating principles, the balance-of-plant, and the safety challenges for airborne hydrogen. Hydrogen fuel cell electric propulsion — using a PEMFC to convert hydrogen and oxygen to electricity, which then drives electric motors — offers the potential for zero-emission flight at ranges extending beyond battery electric capability. The PEMFC operates by electrochemical oxidation of hydrogen at the anode and reduction of oxygen at the cathode, producing water and electricity with no combustion. Understanding the efficiency limit of PEMFCs (approximately 50–60% at practical power densities), the balance-of-plant (the air compressor, humidifier, cooling system, and hydrogen pressure regulator that support the fuel cell stack), and the airborne hydrogen storage challenge (the cryogenic liquid hydrogen or compressed gaseous hydrogen tanks, their insulation requirements, and the specific airworthiness concerns around hydrogen embrittlement, venting, and fire) are the hydrogen propulsion fundamentals that ZeroAvia, Universal Hydrogen, and the Airbus ZEROe programme require at mid-level.

How Electric & Hybrid-Electric Aviation Interviews Differ from Conventional Aerospace and From Automotive/Consumer Electronics Electrical Engineering Roles

- **The weight budget is the governing constraint at every level of the design.** In consumer electronics, an extra 50 grams of battery is a minor packaging inconvenience. In automotive, a 50 kg battery mass increase is significant. In an eVTOL with a 1,800 kg MTOW carrying four passengers, a 50 kg increase in the propulsion system mass is a catastrophic payload reduction. Every design decision in electric aviation — cell chemistry selection, inverter topology, motor winding design, thermal management approach — is evaluated first in terms of its weight consequence. Candidates who come from consumer electronics or automotive electrical engineering without internalising this weight culture make design choices that are technically excellent but operationally disqualifying, and interviewers identify this within the first technical exchange.
- **The certification basis is being developed alongside the aircraft.** Unlike conventional aerospace, where the certification methodology for gas turbine engines and hydraulic actuators has been refined over seventy years of service experience, the certification of an eVTOL battery pack has no established playbook. SC-VTOL-01 provides a framework, but the specific means of compliance — the test data required to demonstrate that a battery thermal runaway event is contained, the operational life limit derivation methodology for a cell chemistry with no aviation service history, the integrity demonstration for a distributed motor array under combined failure modes — are being negotiated between developers and EASA or the FAA as each programme progresses. Senior engineers who have been part of this negotiation — who have written G-1 issue papers, participated in ELOS discussions, or developed novel means of compliance for a component or system without precedent — have an experience that is genuinely unique and specifically valued.
- **Interdisciplinary fluency is tested across wider boundaries than in conventional aerospace.** The eVTOL electrical system engineer sits at the intersection of electrochemistry, power electronics, electric machine design, thermal management, flight control, and structural certification — domains that in conventional aerospace are staffed by separate specialists who communicate through interface documents. In a 50-person eVTOL startup, one engineer may own the BMS, the inverter thermal model, and the motor sizing simultaneously. The interview assesses whether the candidate can engage with the physics of all three with genuine competence or only superficially. The candidate who can explain why the BMS's state-of-charge estimation error directly affects the available energy margin for the emergency reserve, and why this affects the certification argument under SC-VTOL-01's propulsion system integrity requirements, is demonstrating the cross-domain integration that the sector requires.
- **Novel failure modes require specific safety methodology.** The failure mode profile of an electric propulsion system — thermal runaway propagation in a battery pack, motor demagnetisation at elevated temperature, inverter MOSFET shoot-through due to a gate driver fault, common-mode voltage damage to winding insulation — is entirely different from that of a gas turbine engine. Safety analysis tools developed for conventional aviation (FTA, FMEA, Common Cause Analysis) must be applied to these new failure modes with new failure rate data that does not yet exist in any handbook. Engineers who have

performed FMEAs on electric propulsion systems — who have thought through the failure modes of a lithium-ion cell, a BMS contactor, a SiC inverter gate driver, and a PMSM winding insulation failure, and who have assessed the aircraft-level consequence of each — are demonstrating the applied safety methodology that every eVTOL developer requires and that is harder to find than any single-domain technical expertise.

- **The automotive and consumer electronics talent pipelines are the primary competition for the OEM aerospace talent pipeline.** eVTOL developers hire extensively from Tesla, Rivian, Lucid, Waymo, and consumer electronics companies — engineers who have built high-voltage battery packs, permanent magnet motor systems, and automotive-grade inverters at scale. These engineers bring real hardware experience, rapid development culture, and production engineering competence that traditional aerospace candidates often lack. The competitive advantage for aerospace-background candidates is in the certification knowledge, the safety methodology, and the understanding of what it takes to get a novel electrical system approved for operation in a safety-critical environment — domains where automotive engineers consistently discover that the complexity is greater than they expected. The strongest eVTOL engineers combine automotive-pace hardware development experience with aviation-rigour safety engineering, and the interview is assessing which side of this combination the candidate is stronger on and whether the gap on the other side is bridgeable.
- **Simulation and hardware-in-the-loop testing are the development tools, not just the validation tools.** Because there is no large fleet of electric aircraft accumulating service hours from which to derive failure rate statistics, every eVTOL developer must build their reliability case from a combination of accelerated life testing, hardware-in-the-loop (HIL) simulation, and detailed physics-based modelling of the degradation mechanisms. Engineers who have designed and executed an accelerated life test on a battery cell or pack — who have selected the stress factors (temperature, C-rate, depth of discharge) to accelerate the relevant failure modes, collected the data, fitted a degradation model, and used the model to project the field reliability at normal operating conditions — are demonstrating a specific and rare competency that is immediately valuable to every eVTOL developer's certification programme.

PART II: 20 SPECIFIC PREP TASKS

INTERN / GRADUATE

1. Build a complete electric propulsion system energy budget for a specified eVTOL mission 5–6 hours

Skill: The systems-level quantitative reasoning that is the entry-level competency of electric aviation

Why the field cares: The energy budget — computing the energy required for a specified mission from the power required at each flight phase, the duration of each phase, the system losses at each conversion stage, and the required reserves — is the foundational sizing calculation of electric aviation. It is analogous to the rocket propulsion engineer's application of the Tsiolkovsky equation or the aircraft performance engineer's takeoff field length calculation: a first-principles computation that connects the mission requirement to the hardware specification, and that every engineer in the field must be able to perform without reference to a software tool. A candidate who cannot work through this calculation — who cannot estimate battery mass from mission energy, specific energy, and a realistic depth of discharge limit — is demonstrating a gap in the quantitative foundation that the entire field is built on.

Done signal: You can compute, from first principles with explicit assumptions stated, the required battery mass for a specified eVTOL mission scenario — for example: four passengers, 15-minute cruise at 120 kW average power, hover-to-cruise transition and landing at 200 kW peak for 2 minutes each, 20% reserve at end of mission, 80% depth of discharge limit, 85% pack-level efficiency, at three specific energy levels (150, 200, 250 Wh/kg at pack level). You can reason about what the resulting battery mass fraction implies for the maximum takeoff weight allocation between structure, payload, and propulsion, and identify at which specific energy level the mission becomes physically feasible.

5–6 hours

2. Understand lithium-ion cell fundamentals — electrochemistry, C-rate behaviour, ageing mechanisms, and thermal runaway

Skill: The electrochemical foundation of the primary energy storage technology in electric aviation

Why the field cares: Lithium-ion cells are the foundational energy storage technology of electric aviation, and understanding them at the level of their electrochemical mechanisms — not just their headline Wh/kg figures — is the baseline technical competency that every battery engineer, every BMS designer, and every electric propulsion systems engineer must have. The specific interview questions that reveal genuine cell understanding: “why does a lithium-ion cell’s capacity decrease at low temperature?” (the lithium-ion diffusion coefficient in the electrolyte decreases with temperature, increasing internal resistance and causing a larger voltage drop under load), “what causes lithium plating at high C-rate charging?” (lithium ions arrive at the graphite anode faster than they can intercalate, causing metallic lithium deposition on the surface which is a precursor to dendrite growth and thermal runaway), and “what is the thermal runaway cascade sequence for an NMC cell?” (separator melting at $\sim 130^\circ\text{C}$ triggers internal short, electrolyte decomposition at $\sim 200^\circ\text{C}$ releases oxygen, cathode decomposition and electrolyte combustion produce sustained exothermic reaction). These are not academic questions — they are the failure modes that the battery safety case must address.

Done signal: You can explain the lithium-ion intercalation mechanism at both electrodes, describe the role of the SEI (solid-electrolyte interphase) in cell ageing and capacity fade, explain what lithium plating is and why it is accelerated by fast charging at low temperature, describe the thermal runaway cascade sequence for a common cathode chemistry (NMC, LFP, or NCA), and explain why LFP cells are inherently more thermally stable than NMC cells in terms of the oxygen release mechanism. You can sketch the discharge voltage curve of a lithium-ion cell and identify the flat midpoint region and the steep end-of-discharge region, explaining what each represents electrochemically.

4–5 hours

3. Understand the PMSM electric motor fundamentals — torque production, K_e , field weakening, and the efficiency map

Skill: The primary propulsion converter in electric aviation, understood at motor physics depth

Why the field cares: The Permanent Magnet Synchronous Motor is the standard motor type for electric aviation propulsion — it has the best specific power and efficiency of any practical motor architecture at the power levels and speeds relevant to eVTOL. Understanding why it produces torque (the interaction between the rotating stator magnetic field and the rotor permanent magnets), what the back-EMF constant K_e represents (the voltage generated per unit shaft speed, which limits the maximum no-load speed at a given DC bus voltage), why the motor operates in field-weakening at speeds above the base speed (the field-weakening current reduces the effective flux to prevent the back-EMF from exceeding the bus voltage), and what the efficiency map looks like and what its shape implies for the operating strategy during a mission (peak efficiency at a specific torque-speed combination, falling off at low torque and at high speed) are the electric machine fundamentals that motor integration and propulsion system sizing engineers use daily. The candidate who can sketch an approximate efficiency map for a PMSM and explain why the isolines have the shape they do is demonstrating genuine electric machine understanding.

Done signal: You can explain the PMSM torque production mechanism, describe what the back-EMF constant K_e represents and how it limits maximum speed at a given bus voltage, explain what field weakening is and why it is needed for above-base-speed operation, and sketch an approximate PMSM efficiency map — identifying the high-efficiency island, the copper loss-dominated low-speed/high-torque region, and the iron loss-dominated high-speed region. You can explain what the specific power (W/kg) and specific torque (Nm/kg) of the motor tell you about its design, and what limits these values for a permanent magnet machine (thermal limits of the winding insulation and the magnet).

2–3 hours

4. Prepare behavioural answers that demonstrate systems thinking and comfort with novel technical uncertainty

Skill: STAR stories with the novel-problem-under-resource-constraint emphasis eVTOL employers probe

Why the field cares: eVTOL employers — particularly the startup developers — specifically probe how candidates perform when the textbook does not apply, when the test data does not exist, and when the right answer requires building the analytical method from first principles before applying it. The weakest stories in electric aviation interviews end with “I used the standard method to solve the problem.” The strongest end with “there was no established method for characterising the capacity fade of our cell chemistry under the combined stress of high discharge rate and elevated temperature — I designed a design of experiment matrix covering the relevant temperature and C-rate combinations, ran the accelerated life tests, fitted a semi-empirical Arrhenius-based degradation model to the data, and used the model to project the 1,000-cycle capacity retention at operational conditions with a 95% confidence interval, which became the basis for our maintenance interval recommendation.” This story demonstrates experimental design, data fitting, and uncertainty quantification in a context where no standard existed — which is the daily intellectual environment of electric aviation engineering.

Done signal: Four STAR stories covering: a technical problem you solved by developing the method rather than applying an existing one; a cross-disciplinary challenge (an electrical engineering finding with a thermal consequence, or a battery ageing finding with a certification implication); a situation where you worked under resource constraint and had to make a decision based on incomplete data; and a failure in your own analysis or hardware test that you identified and corrected before it reached a design decision. At least two stories involve real electrical hardware or electrochemical data.

5–6 hours

5. Understand power electronics for motor drive — the three-phase inverter topology, SiC versus IGBT switching, and the switching loss mechanism

Skill: The electronic power conversion system that connects the battery to the motor

Why the field cares: The inverter is the component that converts the battery’s DC power to the variable-frequency AC power required by the electric motor, and its efficiency, weight, and reliability are critical to the entire electric propulsion system. The standard topology — the three-phase two-level voltage source inverter with six switches arranged in three half-bridge legs — is the universal architecture for motor drive in electric aviation. Understanding why SiC MOSFETs are preferred over silicon IGBTs for aviation inverters (lower switching losses at high frequency, enabling smaller passive filter components and higher power density), what the switching loss mechanism is (the overlap of current and voltage during the switching transient, multiplied by the switching frequency and summed over the mission), and what the implications are for thermal management (the losses must be conducted away from the semiconductor junction through the package and heatsink to the coolant) are the power electronics fundamentals that every electric propulsion systems engineer must understand. The EMI consequence — the conducted and radiated interference from a kilowatt-scale inverter switching at 20–100 kHz — is a specific and often underestimated certification challenge.

Done signal: You can draw the three-phase two-level VSI topology, identify the six switches and their body diodes, and explain why six complementary switch pairs (not three) are needed for three-phase operation. You can explain the switching loss mechanism — specifically why turn-on loss and turn-off loss are different in a hard-switched inverter — and describe why SiC’s lower gate charge and faster switching speed reduce switching losses relative to silicon IGBT at the same switching frequency. You can explain what the consequence of increasing the switching frequency is for filter size, efficiency, and EMI, and describe what conducted EMI requires in terms of EMI filter design.

6–10 hours

6. Get hands-on with a battery cell characterisation — charge/discharge cycling, internal resistance measurement, and capacity fade tracking

Skill: Practical electrochemical measurement competence as the baseline of battery engineering

6–10 hours

Why the field cares: Battery engineering is fundamentally empirical — cell behaviour is too complex and too material-specific to predict reliably from first principles, and the design of every battery system depends on experimental characterisation of the specific cell chemistry being used. A candidate who has characterised a real lithium-ion cell — performed a constant-current constant-voltage charge cycle, measured the open-circuit voltage as a function of state of charge, measured the DC internal resistance using a pulse technique, and tracked the capacity fade over multiple charge-discharge cycles — has a practical understanding of cell behaviour that is immediately apparent in the specificity with which they discuss battery system design. Accessible cell characterisers (Digatron, Maccor, Biologic) are available at most university engineering departments; lower-cost alternatives (the Powerwall-level charger-discharger combinations, the open-source Battor project) provide accessible entry points for candidates without laboratory access.

Done signal: You have characterised at least one 18650 or pouch cell through a standard test protocol — a formation cycle, a capacity check at 1C, an internal resistance measurement using a 10-second pulse technique, and at least 20 charge-discharge cycles at a specified C-rate and temperature. You can plot the capacity retention versus cycle number, identify any anomalies in the data (an unexpected capacity increase in early cycles due to electrolyte wetting, a sudden capacity decrease suggesting lithium plating or separator damage), and explain what the internal resistance trend tells you about the SEI growth rate. You can describe how the same cell would behave differently if cycled at 2C versus 0.5C.

MID-LEVEL (~3–8 years)

All graduate tasks still apply. Stories must come from professional contexts involving real hardware, real cells, or real certification programmes.

5–6 hours

7. Build thorough working knowledge of Battery Management System design — state estimation, balancing, and the airworthiness safety architecture

Skill: The electronic control system that makes a battery pack safe and operationally useful

Why the field cares: The BMS is the safety-critical electronic system that monitors and controls the battery pack — it is the last line of defence between a cell overcharge or over-temperature event and a thermal runaway. For aviation applications, the BMS safety architecture must be designed so that no single BMS failure can lead to a loss-of-propulsion event above the probability threshold required by SC-VTOL-01 for the failure condition severity, and no single BMS failure can lead to a thermal runaway propagation event that the aircraft cannot survive. This requires redundant measurement channels, independently powered protection circuits, independent contactors for the main and pre-charge circuits, and a fault detection and isolation logic that correctly identifies the BMS's own faults versus pack-level faults. State-of-charge estimation accuracy — the primary operational output of the BMS — directly affects the energy margin available for the mission reserve, and a 5% SOC estimation error at the end of a 15-minute mission can mean the difference between a comfortable reserve and an emergency landing.

Done signal: You can describe the BMS architecture for an aviation battery pack — the measurement channels (cell voltage, pack current, cell temperature at multiple points), the state estimation algorithm (Coulomb counting with OCV correction or extended Kalman filter), the protection functions and their trip points, and the contactor control logic including pre-charge. You can explain why Kalman-filter-based SOC estimation is preferred over pure Coulomb counting for aviation applications (drift correction, noise filtering, ability to fuse multiple measurement types). You can describe the safety architecture requirements for the BMS under SC-VTOL-01 — specifically what redundancy is needed in the measurement and protection paths to meet the single-failure safety requirement for the propulsion system.

5–6 hours

8. Understand SC-VTOL-01 certification framework — the failure classification, propulsion integrity, and current ELOS challenges

5–6 hours

Skill: The regulatory framework within which every eVTOL developer in Europe is certifying their aircraft

Why the field cares: SC-VTOL-01 is the governing certification document for VTOL aircraft in Europe, and understanding it at the level of its specific requirements — not just its existence — is what distinguishes engineers who can contribute to a certification programme from those who cannot. The failure condition classification in SC-VTOL-01 differs from CS-25 in one crucial respect: it explicitly acknowledges that distributed electric propulsion architectures have inherent redundancy, and defines the failure condition severity based on the worst-case effect on the aircraft given the architecture's degraded-mode capability. CS-VTOL-SC.2305 (propulsion system) requires that the loss of any single propulsor leaves the aircraft capable of controlled flight and landing. CS-VTOL-SC.2500 (aircraft systems) requires the systems safety assessment to address all relevant failure conditions including those specific to novel electrical systems. The Equivalent Level of Safety (ELOS) process — through which developers demonstrate that a novel means of compliance achieves a safety level equivalent to the standard requirement — is the mechanism through which most novel battery and propulsion system certification arguments are progressed.

Done signal: You can describe what SC-VTOL-01's Category Enhanced (CAT ENH) classification requires for the loss of a single propulsor, how this differs from the CS-25 requirement for a twin-engine aircraft losing one engine, and what "controlled flight" means in the context of an aircraft with eight motors losing one. You can describe what an ELOS finding is, when it is needed, and what the EASA process for obtaining one involves. You can explain what the specific means of compliance challenges are for an aviation battery pack — specifically the absence of in-service failure rate data for the cell chemistry, the thermal runaway containment demonstration, and the state-of-health monitoring accuracy requirement.

2 hours

9. Prepare a technically specific story about a battery or motor system design problem — requirements, trade-offs, hardware result, and lesson

Skill: Applied electric propulsion engineering at professional depth

Why the field cares: Electric aviation engineering interviews are anchored in the candidate's experience with real hardware — not simulation alone. The most valued stories are those where the hardware behaved differently from the prediction: the battery pack that ran hotter than the thermal model predicted because the inter-cell thermal resistance was higher than the data sheet value; the motor that vibrated unexpectedly because a rotor eccentricity caused a magnetic imbalance that the static finite element model had not captured; the inverter that tripped a protection fault at the worst moment in the mission profile because the DC bus voltage dip under the peak power load was larger than the inverter's undervoltage threshold. These stories reveal both the physical intuition of the engineer and their habit of comparing predictions against measurements — the empirical rigour that the field's nascent reliability database demands.

Done signal: You have a story about a battery, motor, or power electronics system problem from professional or serious research experience — naming the system, the design requirement, the approach taken, the hardware test result that differed from the prediction, the investigation that identified the cause (a modelling assumption, a material data error, an installation effect, a failure mode not in the original FMEA), and the design change that resolved it. The story includes specific numerical values — the predicted versus measured temperature, efficiency, or current draw — and ends with a clear engineering lesson that changed how you approached the next design.

5–6 hours

10. Understand thermal runaway propagation management — the containment requirement, the test methods, and the current state of EASA guidance

Skill: The battery safety engineering challenge at the heart of every eVTOL certification programme

Why the field cares: Thermal runaway in a lithium-ion cell — the exothermic decomposition cascade triggered by internal short circuit, overcharge, or overtemperature — is the most dangerous failure mode in an electric aviation battery pack, and demonstrating that its consequences are contained

5–6 hours

within the pack without causing a catastrophic aircraft-level event is the most technically challenging certification requirement facing every eVTOL developer. The physical challenge: thermal runaway in one cell produces temperatures exceeding 800°C and vents a mixture of flammable gases; the adjacent cells are at risk of runaway propagation through thermal conduction and gas ignition; and the battery management system’s protection functions may be rendered ineffective by the event itself. The engineering responses — cell spacing and thermal barriers to slow conduction, intumescent materials to limit gas propagation, pressure relief venting to prevent mechanical failure, and fire suppression to address ignition — must be demonstrated through a combination of hardware tests (single cell abuse tests, cell-to-cell propagation tests, full-pack thermal runaway tests) and analysis. EASA’s guidance material for the demonstration methodology is still evolving, and the means of compliance for this requirement is one of the most actively negotiated aspects of current eVTOL certification programmes.

Done signal: You can describe the thermal runaway cascade sequence for an NMC cell — identifying the temperature thresholds for SEI decomposition, separator melting, electrolyte decomposition, and cathode oxygen release. You can describe the cell-to-cell propagation mechanism — thermal conduction through the module structure and gas-phase ignition — and identify what design features slow or interrupt propagation (cell spacing, phase change material, ceramic separators, venting channels). You can describe what the FAA’s RTCA DO-311A standard requires for showing that thermal runaway is non-propagating, and what EASA’s equivalent guidance currently requires for SC-VTOL-01 certification.

4–5 hours

11. Understand distributed electric propulsion control — the propulsion-flight control integration and the motor failure response

Skill: The flight control challenge unique to multi-motor electric aircraft

Why the field cares: Distributed electric propulsion aircraft — with 6, 8, 12, or more individual motors — use differential motor thrust as a primary flight control effector, particularly in the hover and transition flight regimes. This means the propulsion system and the flight control system are not merely interfaced through a throttle command and a propulsive axis force — they are deeply integrated, with the flight control computer sending individual torque or speed commands to each motor controller, and using the resulting thrust differentials to generate roll, pitch, and yaw moments. Understanding this integration — how the control allocation algorithm distributes the required moments across the motor array, what the failure response is when one motor fails (how the remaining motors are reallocated to maintain attitude control), and what the handling qualities implications are of the limited residual control authority after a motor failure — is the propulsion-flight control integration knowledge that distinguishes engineers who have worked on DEP aircraft from those who have studied them. The certification consequence: the single-motor failure case must be demonstrated to result in controllable flight, and the flight test methodology for this demonstration is specific to the aircraft configuration.

Done signal: You can describe how a control allocation algorithm distributes moment demands across a multi-motor array — explaining what the objective function is (minimising motor loading while meeting the moment requirement), what the constraint is (maximum motor torque or speed), and how the algorithm handles a motor that is at its limit. You can explain what happens to the attitude control authority when one of eight motors fails — how the remaining seven motors are reallocated, what the resulting control moment envelope is, and whether the aircraft remains controllable within the SC-VTOL-01 requirement. You can describe what the flight test demonstration of single-motor failure recovery requires in terms of test technique and data recording.

3–4 hours

12. Understand eVTOL aerodynamics in hover and transition — the rotorcraft equivalent, the transition challenge, and the power required curve

Skill: The aerodynamics specific to the flight regimes that distinguish eVTOL from conventional fixed-wing aircraft

3–4 hours

Why the field cares: eVTOL aircraft spend significant portions of their missions in hover and in the transition between hover and forward flight — regimes for which the aerodynamic analysis methods developed for conventional fixed-wing aircraft do not apply and for which the rotorcraft methods developed for helicopters apply only approximately to the novel multi-rotor configurations typical of eVTOL. The hover power required — computed from actuator disk theory as $P = (W/n)^{3/2} / \sqrt{2\rho A}$ for n rotors of disk area A at hover weight W in air of density ρ — is the fundamental constraint that determines whether an aircraft can hover at all and for how long. The transition — the regime between pure hover and pure forward flight where the rotor thrust requirements change and the wing begins to generate lift — is the most aerodynamically complex phase and the phase where the power required reaches its maximum for many tiltrotor and lift-plus-cruise configurations. Understanding the power required curve, why the power required in transition may exceed the hover power for a lift-plus-cruise configuration, and what this implies for the battery sizing are the eVTOL aerodynamics fundamentals that every electric propulsion systems engineer who works on rotor-lift aircraft must understand.

Done signal: You can derive the hover power required from actuator disk theory for a specified set of rotor parameters (disk area, hover weight, air density), explain why hover is the most power-demanding steady flight phase for a pure helicopter, and explain why a lift-plus-cruise eVTOL may have a power maximum in transition rather than in hover. You can sketch the power required versus airspeed curve for a representative eVTOL configuration — identifying the hover power, the transition power hump (if present), and the cruise power — and explain what design parameters most strongly influence each region.

4–6 hours

13. Prepare for a power systems design problem under interview conditions — battery sizing, motor selection, or thermal management

Skill: Structured electric propulsion design under time pressure

Why the field cares: Mid-level and senior electric aviation interviews frequently include a live or take-home design problem. “Size the battery pack and motor system for a two-seater eVTOL trainer with a 30-minute endurance at 60 kW cruise power, 120 kW peak for 3 minutes at takeoff, a 10% reserve, 80% DOD, and a MTOW of 1,200 kg — what cell chemistry would you select, what is the pack configuration, what motor continuous and peak power ratings do you need, and what is the approximate thermal management system heat rejection requirement?” This problem requires working through the energy budget, the peak power requirement (which may determine the cell chemistry and C-rate more than the energy requirement), the motor power and torque sizing, and the thermal load from motor and inverter losses at the peak power condition. The candidate who can work through this in a structured, assumption-explicit manner in under 45 minutes, and who knows where the critical design trade-offs are (energy versus power density in cell selection, motor overload factor, thermal management weight penalty), performs substantially better than one encountering the format for the first time.

Done signal: You can work through a complete eVTOL propulsion system sizing problem in under 45 minutes — computing the required energy and peak power from the mission profile, selecting a cell chemistry that meets both the energy density and the C-rate requirement, specifying the battery pack configuration (series-parallel cell arrangement), sizing the motor for the peak torque and continuous torque requirements, and estimating the thermal management heat rejection requirement from the motor and inverter losses at peak power. You have practised this at least twice for different mission profiles.

SENIOR LEVEL (Principal Engineer, Technical Lead, Chief Engineer)

All mid-level tasks still apply. Certification knowledge must extend to direct experience with EASA or FAA means of compliance development. Hardware experience must include a system that has been tested to near-qualification or qualification level.

4–6 hours

14. Prepare to walk through the complete development and certification demonstration of an electric propulsion component — from requirement through test

Skill: End-to-end electric propulsion certification ownership at the level a senior engineer must demonstrate

Why the field cares: Senior electric aviation engineers at developers and propulsion suppliers are expected to have owned the complete development and certification demonstration of a real component — a battery pack, an inverter, a motor, or an integrated propulsion unit — from the requirements derivation through the design through the hardware test programme through the certification evidence package. This is genuinely rare experience in a sector where most programmes have not yet completed first flight, let alone certification. Engineers who have it are specifically valuable, and the interview specifically probes the depth of this ownership: what was the failure mode that the test programme had not predicted, how was the means of compliance for the novel failure mode negotiated with the authority, and what was the qualification evidence that closed the certification finding? The story must be from real hardware, not simulation.

Done signal: For one electric propulsion component or system you personally owned the development and certification demonstration of, you can discuss without notes: the originating system safety requirement (from the aircraft-level FHA or FMEA), the key design trade-off you made and why, the test programme structure (what each phase was intended to demonstrate), the most significant unexpected finding from testing (a failure mode not in the original FMEA, a performance shortfall, or an unexpected interaction with another system), the means of compliance development or negotiation with the authority that addressed it, and the qualification evidence that closed the certification item. You can quantify the unexpected finding and the design or procedure change that addressed it.

4–5 hours

15. Prepare a technically grounded position on one active frontier in electric aviation — solid-state batteries, hydrogen fuel cell integration, or advanced motor technology

Skill: Forward-looking technical awareness at senior engineering depth

Why the field cares: Electric aviation in 2025–2026 has several genuine technical frontiers where the engineering challenges are real, the solutions are not settled, and the architectural decisions being made now will determine which aircraft become viable and which remain stranded. Solid-state batteries — replacing the liquid electrolyte with a ceramic or polymer solid — offer potentially higher energy density, lower thermal runaway risk (no liquid electrolyte to decompose), and wider temperature range operation, but current solid-state cells have lower power density, higher manufacturing cost, and cycle life limitations that are being actively researched. Hydrogen fuel cell integration for regional aviation — combining a PEMFC with a battery buffer to handle peak power, and cryogenic liquid hydrogen storage — faces the balance-of-plant weight penalty, the hydrogen handling certification challenge, and the absence of hydrogen refuelling infrastructure at airports. High-speed permanent magnet motors operating above 20,000 RPM — which offer higher specific power but require careful bearing design, rotor structural analysis, and EMI management — are an active area of motor technology development for direct-drive small propulsors. Senior engineers who can engage at technical depth with one of these frontiers — identifying the specific engineering problem rather than describing the general opportunity — are demonstrating the forward-looking perspective that senior roles require.

Done signal: You have a prepared 3-minute technical discussion covering one specific engineering challenge on one frontier: the ionic conductivity limitation of ceramic solid electrolytes at room temperature and what the engineering consequence is for cold-weather aviation operation; the dynamic load-sharing control challenge between the fuel cell and battery buffer in a hydrogen-electric propulsion system during the transition from cruise to approach power; or the rotor mechanical integrity challenge for a 20,000 RPM PMSM subjected to the combined thermal, centrifugal, and magnetic load at peak speed. Your position identifies the specific unresolved engineering problem and the approaches being taken, not just the high-level opportunity.

16. Prepare to lead a technical discussion on an electric propulsion architecture trade-off — series hybrid, parallel hybrid, or pure electric for a specified mission 4–5 hours

Skill: Architectural decision-making at senior engineering speed

Why the field cares: Senior electric aviation engineers make or advise on the propulsion architecture decisions that determine whether a programme is technically feasible, commercially viable, and certifiable. “For a 19-passenger regional commuter aircraft targeting 200 nm range with 45-minute reserve — compare pure battery electric, series hybrid with a turbogenerator, and hydrogen fuel cell electric architectures in terms of the key performance metrics, the certification complexity, and the infrastructure dependency.” This requires reasoning through: the specific energy of each energy source, the energy required for the mission, the conversion efficiency chain, the weight penalty of each architecture’s ancillary components, the certification challenges specific to each, and the operational infrastructure requirements. The senior interview tests whether you can structure this reasoning live and arrive at a specific architecture recommendation with explicit trade-offs and stated sensitivities to technology assumptions.

Done signal: You have worked through at least two electric propulsion architecture trade-off problems spanning at least two of: pure battery electric versus series hybrid for a specified range and payload; hydrogen fuel cell electric versus turbogenerator hybrid for a regional commuter mission; distributed electric propulsion with battery versus centralised motor with gearbox for a specific eV-TOL configuration; or battery chemistry selection (NMC vs LFP vs solid-state) for a mission with a specific energy density requirement and a thermal management constraint. For each, you state the governing constraint, the key trade-off, and the architecture recommendation with explicit sensitivities to the technology assumptions.

17. Understand the FAA powered-lift certification pathway — how it differs from SC-VTOL-01 and the current FAR Part 21/91/135 rulemaking status 4–5 hours

Skill: The US regulatory framework for eVTOL certification, distinct from the European approach

Why the field cares: The FAA’s certification approach for powered-lift aircraft — aircraft that use lift generated by engine power for vertical takeoff and landing but that rely on non-rotating surfaces for forward flight — has evolved significantly since 2020. The FAA issued a final rule in October 2023 designating powered-lift as a new aircraft category, establishing airworthiness standards under Part 21, and creating operational rules under a new Part 135 amendment. FAA Order 8110.56A provides guidance on the certification process for powered-lift type certificates. The standards applicable to powered-lift are based on CS-25 / FAR Part 25 for large aircraft and CS-27 / FAR Part 27 for small rotorcraft, with Special Conditions addressing novel features — a different approach from EASA’s purpose-built SC-VTOL-01. Understanding the US pathway — specifically the powered-lift category definition, the applicability of airworthiness standards, and the operational rule requirements for Part 135 commercial operations — is the regulatory knowledge required for engineers working on programmes targeting the US market (Joby, Archer, Wisk, Beta Technologies are all certifying under the FAA pathway).

Done signal: You can describe the FAA’s powered-lift category as established in the 2023 final rule — what the category definition is, how it differs from airplane and rotorcraft categories, and what airworthiness standards apply. You can explain how the FAA’s Special Conditions approach to novel features differs from EASA’s SC-VTOL-01 approach, and describe at least two specific areas where the two authorities’ approaches diverge in their requirements for the electric propulsion system. You can describe the current status of the FAA Part 135 operational rules for powered-lift commercial passenger operations.

18. Build quantitative estimation fluency for electric propulsion performance — power, energy, thermal load, and system mass without simulation 4–5 hours

Skill: Senior-speed first-principles reasoning about electric propulsion system feasibility

Why the field cares: Senior electric aviation engineers are expected to sanity-check system designs

4–5 hours

from junior engineers, estimate whether a proposed propulsion architecture is feasible before committing to a design trade study, and recognise physically implausible performance claims from suppliers or partners. The estimation problems are tractable from first principles: estimating the hover power from actuator disk theory; estimating the battery mass from the mission energy and specific energy; estimating the motor loss at a specified operating point from the winding resistance and iron loss coefficients; estimating the inverter heat rejection requirement from the switching loss at a specified switching frequency and current; and estimating the thermal management system mass from the heat rejection requirement and a specific coolant flow rate. None of these requires a simulation — they require the governing physical relationships and engineering judgment about typical parameter values.

Done signal: You have worked through five electric propulsion estimation problems covering at least two of: hover power from actuator disk theory for a specified rotor configuration; battery mass from mission energy and specific energy with reserve margin; motor copper loss from winding resistance and peak current; inverter switching loss from device data sheet parameters at a specified switching frequency; and thermal management system coolant flow rate from heat rejection requirement and temperature rise limit. You state your assumptions before numbers, give order-of-magnitude answers with stated uncertainty, and identify the most load-bearing assumption in each.

ALL LEVELS

1–2 hours

19. Prepare a specific answer to "Why electric aviation — and why this specific technology area?" that reflects genuine technical engagement

Skill: Technically specific motivation that distinguishes electric aviation-committed candidates from those applying broadly to the sector

Why the field cares: Electric aviation employers — particularly the startup developers — are looking for engineers who are genuinely engaged with the specific technical challenges of the field, not simply attracted to its novelty or its sustainability narrative. The expected answer in 2025–2026 engages with one of: the battery specific energy challenge and what solid-state or lithium-sulphur chemistries need to achieve before specific mission profiles become viable; the thermal runaway containment certification challenge and what the engineering approaches for multi-cell propagation prevention are; the distributed electric propulsion control problem and what the control allocation and failure response methodology requires; the hydrogen fuel cell balance-of-plant weight problem for regional aviation; or the specific certification methodology challenge for a novel component with no in-service failure rate database. The answer must name one specific engineering challenge and connect the candidate's background to contributing to its solution, not just to the general mission.

Done signal: You have a 90-second answer that names your target electric aviation domain (battery systems, power electronics, motor design, thermal management, systems integration, or certification methodology), identifies one specific active engineering challenge in that domain, explains what makes it technically hard (the specific physics or regulatory constraint), and connects your engineering background to contributing to its solution. The answer cannot be given unchanged to a candidate for a general electrical engineering or automotive powertrain role.

1–2 hours

20. Map your technical background to a specific employer type, technology domain, and development stage — and understand where the sector is in 2025

Skill: Precise self-positioning in a sector whose employment landscape is changing rapidly

Why the field cares: The electric aviation sector in 2025 is in a critical transition — from the development and certification phase to the early commercial operations phase, with the first eVTOL type certificates being issued, the first commercial routes being operated, and the first production programmes scaling up. This transition changes what employers need: development-phase employers need engineers who can work through ambiguous technical problems without established handbook methods; production-phase employers need engineers who can translate the develop-

1–2 hours

ment programme’s design decisions into production-ready manufacturing and quality processes. A candidate who does not understand where a specific employer is in this transition — who applies for a production ramp-up role with a development-phase mindset, or vice versa — is mismatching their experience to the employer’s need. Similarly, a candidate who applies to Joby (FAA certification phase, 2025) without having studied the publicly available FAA certification engagement documents, or to Vertical Aerospace (EASA SC-VTOL-01 pathway) without understanding SC-VTOL-01, is demonstrating a preparation gap that these employers notice immediately.

Done signal: You can state in under 60 seconds: the electric aviation employer type you are targeting (pure-play eVTOL developer, hybrid-electric regional developer, incumbent OEM electric programme, propulsion supplier, or research institution), the specific technology domain you are most experienced in, the development stage of the programmes you have worked on (concept, preliminary design, detailed design, testing, or certification), the programming and simulation tools you work in fluently, and two specific technical contributions your background makes to the target role. You have researched the specific employer’s current certification status and public programme milestones, and you can describe where they are in their development programme and what technical challenges they are currently facing.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Electric propulsion energy budget — mission profile to battery mass, three specific energy levels	All	5–6 h
2	Lithium-ion cell fundamentals — electrochemistry, C-rate, ageing, thermal runaway cascade	All	5–6 h
3	PMSM motor fundamentals — torque production, K_e , field weakening, efficiency map	All	4–5 h
4	STAR stories — novel problem, cross-disciplinary, resource constraint, hardware failure	All	2–3 h
5	Power electronics — three-phase VSI, SiC vs IGBT, switching loss mechanism, EMI	All	5–6 h
6	Battery cell characterisation — cycling, internal resistance, capacity fade tracking	All	6–10 h
7	BMS design — state estimation, balancing, airworthiness safety architecture	Mid+	5–6 h
8	SC-VTOL-01 certification framework — failure classification, propulsion integrity, ELOS	Mid+	5–6 h
9	Battery/motor system story — requirements, trade-offs, hardware result, lesson	Mid+	2 h
10	Thermal runaway propagation management — cascade, containment, test methods, EASA guidance	Mid+	5–6 h
11	DEP control — propulsion-FCS integration, failure response, control allocation	Mid+	4–5 h
12	eVTOL aerodynamics — hover power, transition, power required curve	Mid+	3–4 h
13	Propulsion design problem under time pressure — battery sizing, motor, thermal management	Mid+	4–6 h
14	Complete propulsion certification story — requirement through qualification test	Senior	4–6 h
15	One active frontier — solid-state batteries, hydrogen fuel cell, or high-speed motors	Senior	4–5 h
16	Architecture trade-off — series hybrid, parallel hybrid, pure electric for specified mission	Senior	4–5 h
17	FAA powered-lift certification pathway — category definition, standards, Part 135 operations	Senior	4–5 h
18	Quantitative estimation — hover power, battery mass, motor loss, thermal load	Senior	4–5 h
19	"Why electric aviation — why this technology domain?" — technically specific	All	1–2 h
20	Employer type / technology domain / development stage / current sector status	All	1–2 h

THE STRUCTURAL REALITY THAT MAKES ELECTRIC & HYBRID-ELECTRIC AVIATION DIFFERENT

Electric and hybrid-electric aviation is the only aerospace subdomain where the most critical engineering challenges are simultaneously physical, electrochemical, regulatory, and commercial — and where the solution to any one of them is contingent on progress in the others. The battery energy density that makes a 200-nm electric regional aircraft viable does not yet exist at commercially manufacturable cost. The regulatory framework that would certify a novel solid-state battery chemistry in an airborne application does not yet have established precedent. The production volume that would bring the unit cost of electric propulsion components down to commercial aviation's tolerance is not achievable until the certification is complete. And the certification cannot be completed without the production-representative hardware that only volume manufacturing provides.

This circularity — unique to electric aviation — creates an engineering environment unlike any other in aerospace. The engineers working in this sector are simultaneously designing hardware, developing the test methods by which that hardware will be qualified, negotiating the regulatory standards against which it will be certified, and modelling the electrochemical degradation processes that will determine its operational life before the first commercial operation has accumulated enough hours to validate any of these models. Every assumption has larger uncertainty than in any mature aerospace discipline. Every design decision is made with data that is incomplete by the standards the decision warrants.

The engineers who succeed in this environment are those who are simultaneously rigorous about the physics they can characterise precisely — the thermodynamics of the motor losses, the electrokinetics of the cell's internal resistance — and appropriately humble about the phenomena they cannot yet characterise reliably — the long-term degradation of a novel cell chemistry under the specific thermal and mechanical loading of an airborne application, or the certification authority's response to a means of compliance they have never evaluated before. They use the data they have aggressively and the data they do not have conservatively. They close the analysis where they can and flag the uncertainty where they cannot. And they understand that in a sector where the certificate takes years to obtain and the failure of any individual programme sets back the entire sector's credibility with the authority, the conservative decision made from incomplete data is almost always better than the aggressive decision made from the same data.

The opportunity, for engineers who can navigate this combination of rigour and uncertainty, is genuine: the engineering problems are unsolved, the market is real, the regulatory engagement is direct rather than mediated through decades of established precedent, and the contribution of any individual engineer to the first commercially viable electric aircraft is traceable and lasting in a way that contribution to a mature aviation programme rarely is.